

FHIR Task Workflow Orchestration Guide

Overview

This document explains how to use the FHIR Task resource as the orchestration engine for hospital workflows.

This is one of the most important resources for enterprise-grade systems.

Without Task:

- Workflow tracking becomes difficult
- Queue management becomes weak
- Multi-department coordination becomes messy
- AI workflow orchestration becomes hard
- Automation pipelines become fragmented

With Task:

You can build:

- Queue systems
- AI workflows
- Hospital operations pipelines
- Multi-step workflows
- BPM engines
- Workflow automation
- Human approval systems
- Async clinical workflows

Why Task is Important

FHIR resources like:

- Observation
- Condition
- MedicationRequest
- Procedure
- ServiceRequest

store clinical data.

But they DO NOT manage workflow state.

Example:

A ServiceRequest says:

"Do CBC test"

But it does not track:

- waiting for sample
- sample collected
- sample in lab
- processing
- verified
- completed

Task solves this.

What Task Represents

Task represents:

A unit of work

Examples:

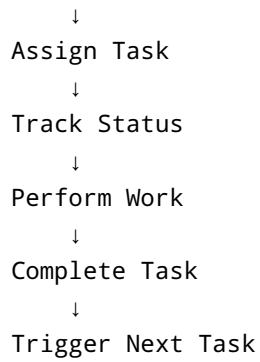
- Collect blood sample
 - Call patient
 - Review lab report
 - Approve insurance
 - Dispense medication
 - Start consultation
 - Move patient to room
 - Upload report
 - AI summarize encounter
-

Recommended Task Architecture

Business Event

↓

Create Task



Recommended Task Status Lifecycle

Use these statuses consistently.

Status	Meaning
requested	Created but not accepted
ready	Ready to start
in-progress	Currently executing
on-hold	Paused
completed	Finished
failed	Failed
cancelled	Cancelled

Recommended Task Categories

You should create internal workflow categories.

Category	Purpose
registration	Patient registration
triage	Nursing/vitals
consultation	Doctor consultation
laboratory	Lab workflows

Category	Purpose
radiology	Imaging workflows
pharmacy	Medication dispensing
billing	Billing operations
insurance	Claim processing
ai	AI-generated workflows
admin	Administrative operations

CORE TASK RELATIONSHIPS

Task

- |— Patient
- |— Encounter
- |— Appointment
- |— ServiceRequest
- |— Procedure
- |— DiagnosticReport
- |— MedicationRequest
- |— Practitioner
- |— Organization

OUTPATIENT TASK FLOW

High-Level OP Flow

```

Appointment Booked
  ↓
Task: Check-in Pending
  ↓
Task: Collect Vitals
  ↓
Task: Doctor Consultation
  ↓
Task: Lab Collection
  ↓

```

```
Task: Lab Processing
  ↓
Task: Doctor Review
  ↓
Task: Pharmacy Processing
  ↓
Task: Billing
  ↓
Task: Insurance Processing
  ↓
Task: Encounter Completion
```

STEP 1 — Appointment Booked

Appointment gets created.

Appointment Example

```
{
  "resourceType": "Appointment",
  "id": "appointment-1",
  "status": "booked",
  "participant": [
    {
      "actor": {
        "reference": "Patient/patient-1"
      },
      "status": "accepted"
    }
  ]
}
```

STEP 2 — Create Check-in Task

Purpose

Front desk should check in patient.

Task Example

```
{
  "resourceType": "Task",
  "id": "task-checkin-1",
  "status": "requested",
  "intent": "order",
  "code": {
    "text": "Patient Check-in"
  },
  "for": {
    "reference": "Patient/patient-1"
  },
  "focus": {
    "reference": "Appointment/appointment-1"
  },
  "owner": {
    "reference": "Organization/frontdesk"
  }
}
```

STEP 3 — Check-in Completed

Task becomes:

```
"status": "completed"
```

Then system creates:

- Encounter
- Next task

STEP 4 — Create Encounter

Example

```
{
  "resourceType": "Encounter",
  "id": "encounter-1",
```

```
"status": "in-progress",
"subject": {
  "reference": "Patient/patient-1"
}
}
```

STEP 5 — Create Vitals Task

Purpose

Nurse collects vitals.

Example

```
{
  "resourceType": "Task",
  "id": "task-vitals-1",
  "status": "ready",
  "intent": "order",
  "code": {
    "text": "Collect Vitals"
  },
  "for": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  },
  "owner": {
    "reference": "Practitioner/nurse-1"
  }
}
```

STEP 6 — Nurse Records Vitals

Observation Example

```
{
  "resourceType": "Observation",
```

```
"id": "obs-temp-1",
"status": "final",
"code": {
  "text": "Body Temperature"
},
"subject": {
  "reference": "Patient/patient-1"
},
"encounter": {
  "reference": "Encounter/encounter-1"
},
"valueQuantity": {
  "value": 98.6,
  "unit": "F"
}
}
```

STEP 7 — Complete Vitals Task

Task status:

```
"status": "completed"
```

System triggers next task.

STEP 8 — Doctor Consultation Task

Example

```
{
  "resourceType": "Task",
  "id": "task-consult-1",
  "status": "ready",
  "intent": "order",
  "code": {
    "text": "Doctor Consultation"
  },
  "for": {
    "reference": "Patient/patient-1"
  },
}
```



```
"encounter": {
  "reference": "Encounter/encounter-1"
},
"owner": {
  "reference": "Practitioner/prac-1"
}
}
```

STEP 9 — Doctor Orders Lab

Doctor creates ServiceRequest.

Example

```
{
  "resourceType": "ServiceRequest",
  "id": "sr-cbc-1",
  "status": "active",
  "intent": "order",
  "code": {
    "text": "Complete Blood Count"
  },
  "subject": {
    "reference": "Patient/patient-1"
  }
}
```

STEP 10 — Create Lab Collection Task

This is where Task becomes powerful.

Example

```
{
  "resourceType": "Task",
  "id": "task-lab-collect-1",
  "status": "requested",
  "intent": "order",
  "code": {
```

```
    "text": "Collect Blood Sample"
  },
  "for": {
    "reference": "Patient/patient-1"
  },
  "focus": {
    "reference": "ServiceRequest/sr-cbc-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-1"
  },
  "owner": {
    "reference": "Organization/laboratory"
  }
}
```

STEP 11 — Sample Collection Workflow

Workflow States

```
requested
  ↓
ready
  ↓
in-progress
  ↓
completed
```

STEP 12 — Create Lab Processing Task

Once sample collected:

Task: Lab Processing

Example

```
{
  "resourceType": "Task",
```

```
"id": "task-lab-process-1",
"status": "ready",
"intent": "order",
"code": {
  "text": "Process CBC"
},
"focus": {
  "reference": "ServiceRequest/sr-cbc-1"
},
"for": {
  "reference": "Patient/patient-1"
}
}
```

STEP 13 — Diagnostic Result Generation

Lab generates:

- Observation
- DiagnosticReport

DiagnosticReport Example

```
{
  "resourceType": "DiagnosticReport",
  "id": "dr-cbc-1",
  "status": "final",
  "subject": {
    "reference": "Patient/patient-1"
  },
  "basedOn": [
    {
      "reference": "ServiceRequest/sr-cbc-1"
    }
  ]
}
```

STEP 14 — Create Doctor Review Task

Purpose

Doctor reviews report.

Example

```
{
  "resourceType": "Task",
  "id": "task-review-1",
  "status": "ready",
  "intent": "order",
  "code": {
    "text": "Review Lab Result"
  },
  "focus": {
    "reference": "DiagnosticReport/dr-cbc-1"
  },
  "owner": {
    "reference": "Practitioner/prac-1"
  }
}
```

STEP 15 — Prescription Workflow

Doctor prescribes medication.

MedicationRequest Example

```
{
  "resourceType": "MedicationRequest",
  "id": "medreq-1",
  "status": "active",
  "intent": "order",
  "subject": {
    "reference": "Patient/patient-1"
  },
  "medicationCodeableConcept": {
    "text": "Paracetamol 500mg"
  }
}
```

```
}  
}
```

STEP 16 — Create Pharmacy Task

Example

```
{  
  "resourceType": "Task",  
  "id": "task-pharmacy-1",  
  "status": "requested",  
  "intent": "order",  
  "code": {  
    "text": "Dispense Medication"  
  },  
  "focus": {  
    "reference": "MedicationRequest/medreq-1"  
  },  
  "for": {  
    "reference": "Patient/patient-1"  
  },  
  "owner": {  
    "reference": "Organization/pharmacy"  
  }  
}
```

STEP 17 — Billing Task

Purpose

Billing team prepares invoice.

Example

```
{  
  "resourceType": "Task",  
  "id": "task-billing-1",  
  "status": "ready",  
  "intent": "order",  
}
```

```
"code": {
  "text": "Generate Invoice"
},
"for": {
  "reference": "Patient/patient-1"
},
"encounter": {
  "reference": "Encounter/encounter-1"
}
}
```

STEP 18 — Invoice Generation

Invoice Example

```
{
  "resourceType": "Invoice",
  "id": "invoice-1",
  "status": "issued",
  "subject": {
    "reference": "Patient/patient-1"
  },
  "totalNet": {
    "value": 4500,
    "currency": "INR"
  }
}
```

STEP 19 — Insurance Workflow

Claim Example

```
{
  "resourceType": "Claim",
  "id": "claim-1",
  "status": "active",
  "patient": {
    "reference": "Patient/patient-1"
  }
}
```

```
}  
}
```

STEP 20 — Insurance Approval Task

Example

```
{  
  "resourceType": "Task",  
  "id": "task-insurance-1",  
  "status": "requested",  
  "intent": "order",  
  "code": {  
    "text": "Insurance Approval"  
  },  
  "focus": {  
    "reference": "Claim/claim-1"  
  }  
}
```

TASK CHAINING

One of the strongest patterns.

```
Task A completes  
  ↓  
Create Task B  
  ↓  
Task B completes  
  ↓  
Create Task C
```

Recommended Event-Driven Architecture

Recommended Events

```
task.created  
task.assigned  
task.started  
task.completed  
task.failed
```

Clinical events:

```
appointment.booked  
encounter.started  
service_request.created  
diagnostic_report.completed  
invoice.generated  
claim.submitted
```

AI WORKFLOW INTEGRATION

Task is PERFECT for AI orchestration.

Example AI Workflow

```
Encounter Completed  
↓  
AI Task Created  
↓  
Generate Encounter Summary  
↓  
Generate Coding Suggestions  
↓  
Generate Claim Draft  
↓  
Human Review Task
```


AI Task Example

```
{
  "resourceType": "Task",
  "id": "task-ai-summary-1",
  "status": "requested",
  "intent": "order",
  "code": {
    "text": "AI Generate Encounter Summary"
  },
  "focus": {
    "reference": "Encounter/encounter-1"
  },
  "owner": {
    "display": "AI Workflow Engine"
  }
}
```

QUEUE MANAGEMENT WITH TASK

Task can power:

- doctor queue
- nursing queue
- pharmacy queue
- lab queue
- billing queue

Example Queue Query

```
GET /Task?status=ready&code=Collect Vitals
```

Returns:

```
Patients waiting for vitals
```

PRIORITY MANAGEMENT

Task supports priority.

Example

```
"priority": "urgent"
```

Useful for:

- STAT labs
- Emergency review
- Critical findings

ESCALATION WORKFLOWS

Example:

```
Task not completed within 30 mins
↓
Escalate to supervisor
↓
Create escalation task
```

TASK OWNERSHIP MODELS

1. Human Assignment

```
Task → Practitioner
```

Example:

```
Assigned to Dr. Kumar
```

2. Department Assignment

Task → Organization

Example:

Assigned to Laboratory

3. AI Assignment

Task → AI Agent

Example:

AI summarization

RECOMMENDED TASK SEARCHES

Tasks for Patient

GET /Task?patient=Patient/patient-1

Tasks for Encounter

GET /Task?encounter=Encounter/encounter-1

Pending Tasks

```
GET /Task?status=requested
```

Department Queue

```
GET /Task?owner=Organization/laboratory
```

Recommended Production Architecture

Suggested Modules

```
workflow/  
  task_engine  
  task_rules  
  queue_management  
  escalation_engine  
  automation_engine  
  ai_orchestration
```

Suggested Internal Workflow Engine

```
Event Bus  
  ↓  
Workflow Rules Engine  
  ↓  
Task Generator  
  ↓  
Task Queue  
  ↓  
Notification Engine
```

RECOMMENDED DATABASE TABLES

Core Tables

```
task
task_history
task_assignment
task_comment
task_event
task_queue
```

Recommended Task Metadata

Useful additional fields:

Field	Purpose
queue_number	Queue management
sla_due_time	Escalation/SLA
retry_count	Retry workflows
workflow_stage	Tracking
automation_source	AI/manual/system
parent_task_id	Task hierarchy

TASK VS SERVICE REQUEST

Very important distinction.

ServiceRequest	Task
Clinical order	Workflow action
"Do CBC"	"Collect sample"
Medical intent	Operational execution
Clinical domain	Workflow domain

TASK VS APPOINTMENT

Appointment	Task
Scheduled visit	Action item
Time reservation	Unit of work
Scheduling	Operations

TASK HIERARCHY EXAMPLE

```
Encounter Workflow Task
├── Vitals Task
├── Consultation Task
├── Lab Task
│   ├── Sample Collection
│   ├── Processing
│   └── Verification
├── Pharmacy Task
└── Billing Task
```

FINAL RECOMMENDED RESOURCES FOR WORKFLOW ENGINE

Strongly recommended set:

```
Task
Appointment
Encounter
ServiceRequest
DiagnosticReport
Procedure
MedicationRequest
Observation
Claim
Invoice
Practitioner
```

Organization
Location

FINAL ARCHITECTURE RECOMMENDATION

If you want enterprise-grade FHIR workflows:

Use:

FHIR Resources = Clinical Data Layer
Task Resource = Workflow Layer
Event Bus = Integration Layer
AI Agents = Automation Layer

This architecture scales VERY well.

FINAL SUMMARY

Task is the missing orchestration layer between:

FHIR Clinical Data
AND
Hospital Operations

Task enables:

- workflow orchestration
- queues
- automation
- AI workflows
- async processing
- escalations
- BPM engines
- human approvals
- operational tracking

Without Task:

FHIR becomes mostly a clinical storage model.

With Task:

FHIR becomes a full operational healthcare platform.